Dohyeop Lim

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Education

Seoul National University of Science and Technology 2024 – Present
- *B.E. in Applied Artificial Intelligence*
- GPA 4.34 / 4.5 (Rank: 1/62)

Research Experience

Visual Intelligence Lab, SeoulTech Dec. 2025 – Present
- *Undergraduate Researcher*, Advisor: Prof. Jongyoul Park
- Research on vision-language models and document AI.

Research Projects

Automated Data Preprocessing for Industrial AI (KIAT) Mar. 2026 – Present
- Recruited annotators and coordinated collection and labeling of 8,000 VIN images.
- Developed a real-time image processing and decoding library for integration into KETI and INTERX platforms, achieving sub-17 ms processing latency on a mobile CPU.
- Preprint [1], patent application [P1], and software copyrights [S1, S2].

Distributed GPU-based Vision-Language-Action Model Scaling (IITP) Apr. 2026 – Present
- Assisted with data center operations and GPU resource allocation using MIG and Kubernetes.
- Preprints [1, 2], and software copyrights [S3].

Generative AI for Accessible Content Creation for Seniors (KOCCA) Mar. 2026 – Present
- Participated in research on image generation models with ETRI.

Publications

[1] **Dohyeop Lim**, MinKi Jeong, and Jongyoul Park. "E-ACT: Enforcing Arithmetic Constraints in CTC Trellises for Structured Identifier Recognition." 2026. *Preprint*
- Developed constraint-aware CTC decoding, improving exact-match accuracy by an average of 9.5 percentage points across four identifier domains.

[2] MinKi Jeong, **Dohyeop Lim**, and Jongyoul Park. "Imagine the Structure Before You Speak: Structural Imagination in Latent Space for Table Recognition." 2026. *Preprint*

[3] Bo Sung Kim, Jae Yeop Kim, Sanghyung Jin, **Dohyeop Lim**, Hyojin Lim, and Chae Jin Lim. "A furin-cleavable pro-peptide complex enhances anti-aging efficacy via microneedling-assisted delivery." 2026. *Under review at Cosmetics (JCR Q1)*

Intellectual Property

Patents

[P1] Method and System for Label-Free Self-Adaptive Identifier Recognition Using Multiple Recognizers and Identifier Specification Constraints. *Patent pending*

Software Copyrights

[S1] Method and System for CTC-Based Text Recognition Using Identifier Rules.

[S2] Method and System for Evaluating OCR Recognition Performance and Analyzing Error Changes Across Image Processing Techniques.

[S3] AI Agent Orchestration System for Research.

Honors

Full-Tuition Academic Excellence Scholarship, 5 consecutive semesters 2024 – 2026

Languages

TOEFL iBT: 100 Aug. 2026

Activities

Startup

Collog: Turning calls with family into voice-based health insights. [link] Aug. 2026 – Present
- Selected for the 2026 Modoo Startup Project, 10.8% acceptance rate.

International

Visiting Student, Technische Hochschule Ulm, Germany Jan. – Feb. 2026

Leadership

LIKELION University SeoulTech Chapter, Vice President 2025 – Present

Google Developer Groups on Campus, Core Member 2025 – Present